Fiscal Year:	FY 2005	**Task Last Updated:**	FY 12/13/2004
PI Name:	Tanveer, Saleh Ph.D.		
Project Title:	Dendritic Crystal Growth		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	tanveer@math.ohio-state.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	614-292-5710
Organization Name:	Ohio State University		
PI Address 1:	420 MATH TOWER		
PI Address 2:	231 W 18TH AVE		
PI Web Page:	http://www.math.ohio-state.edu/		
City:	COLUMBUS	**State:**	OH
Zip Code:	43210	**Congressional District:**	15
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Fluid Physics 01-OBPR-02: Research and Flight Experiment Opportunities
Start Date:	01/01/2002	**End Date:**	07/18/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Mohammad, Kassemi	**Contact Phone:**	216-433-5031
Contact Email:	Mohammad.Kassemi@grc.nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 7/18/2006, per N. Hall (9/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Foster, Michael R (The Ohio State University)		
Grant/Contract No.:	NAG3-2700		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to investigate dendritic crystal growth in two and three dimensions, particularly when surface energy is small. The particular goals are: (i) to study steady two and three dimensional dendritic crystal growth with different types of small effects, (ii) to recalculate the selection parameter for a non-axi-symmetric 3D dendrite that accounts for non-axi-symmetric corrections to Ivantsov paraboloid in a consistent manner, (iii) to determine if a singular perturbation of an otherwise smoothly evolving interface obtained before for one-sided 2-D model hold more generally for 3-D axisymmetric case and/or for two sided models. Importance of the study is in predicting operating conditions for a dendrite, and determining if, given the near-structural instability for small-surface-energy anisotropy, there is a cross-over in the dependence of the selection parameter, usually denoted in the literature by ``sigma^* ''. Such a cross-over may serve to explain the absence of experimental evidence of a strong dependence on surface energy anisotropy as it tends to zero, as predicted by microscopic solvability. The time-evolution scenario for singular perturbations that we seek to study will help confirm the dynamical selection mechanism already found by the PI's for a two-dimensional (2D) one-sided model. In particular, it will solidify evidence against the marginal stability hypothesis that is still widely popular. Further, our coarsening result, if confirmed for 3D could set the stage for appropriate experiment to determine if, in the limit of small sigma^*, there is an intermediate range of distances from the dendrite tip where a power law different from a third (predicted in the far-field by Voorhees & Glicksman) is valid.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The general goal is to have a fundamental understanding of the dendritic crystal growth process. This includes prediction of micro-structure scales and what controls it. It has been known for a while that these are controlled by tip features. Therefore, a clear understanding of the tip selection for a realistic 3-D dendrite is an important goal. With respect to this goal, we have re-examined Brenner and Ben Amar theory for 3-D growth crystal growth and have found some inconsistencies in their assumption. Non-axisymmetric description of the tip has been arbitrarily truncated in the Brenner's work to a four-fold symmetry term. It has been implicitly assumed that higher order terms are small. We find otherwise-- that there is substantial correction to microscopic solvability when higher order terms are included and one allows for general background shapes than Ivantsov. This is being written up in a paper with co-PI make Foster. The selection theory itself is in a state of flux. There have been reported numerical calculations with phase field model with no evidence of strong role of anisotropy that one sees in microscopic solvability; others claim just the opposite. There are fundamental questions if the formal asymptotic calculations are actually correct mathematically. This has been tackled in a series of rigorous mathematical papers on microscopic solvability starting with the closely mathematically allied Hele-Shaw problem and continued extension by a former student in the dendritic crystal growth problem. These are significant additions to the literature as they clearly and completely answer the question about the correctness of the underlying steady state microscopic solvability theory from a mathematical view point, at least for the 2-D case. These papers have already come out. Despite this, questions persist about the time-evolving flow about whether or not an actual time-dependent problem evolves in a way consistent with a steady tip theory. With respect to this goal, we are doing a combined analytical and numerical project with a student to see how disturbances on a background shape evolve nonlinearly as they advect on a dendrite for both 2-D and 3-D without assuming the underlying shape to be Ivantsov.
Bibliography Type:	Description: (Last Updated: 03/20/2017)
Articles in Peer-reviewed Journals	Xie X, Tanveer S. "Rigorous results in steady finger selection in viscous fingering." Archive for Rational Mechanics and Analysis, 2003 Feb;166(3):219-86. , Feb-2003
Articles in Peer-reviewed Journals	Tanveer S, Xie X. "Analyticity and nonexistence of classical steady Hele-Shaw fingers," Communications on Pure and Applied Mathematics 2003 Mar;56(3):353-402. , Mar-2003
Articles in Peer-reviewed Journals	Xie X. "Rigorous results in selection of steady needle crystals." Journal of Differential Equations 2004 Mar 1;197(2):349-426. , Mar-2004
Presentation	Foster, M.R.; Tanveer, S. "Tip selection in three-dimensional dendrites " American Physical Society Division of Fluid Dynamics Annual Meeting Nov-2004